**Input: a truth table (one K-map per output $f_1, \ldots, f_m$)**

The cells where an output is 1 form its *on-set*. A quantum circuit must *flip* an output wire (initially $|0\rangle$) exactly for those input rows.

*Example, full adder: inputs $A, B, C_{in}$; outputs Sum (odd parity) and Carry (majority).*

## This work: disjoint K-map groups

**1. Disjoint cube cover (Algorithm 1)**

Grow each K-map group (a cube of $2^d$ cells) only inside cells not yet covered, so no two groups overlap.

*Sum: 4 single cells. Carry: group $BC_{in}$ plus 2 single cells.*

**2. Each group → one multi-controlled $X$**

Controls on the group's fixed inputs; an input that must be 0 is wrapped in $X$ gates. The groups are disjoint, so each output flips exactly once per 1-cell (Theorem 2).

*Full adder: 23 gates.*

**3. Derivation (Lemma 1)**

Build a later output as a CNOT from an earlier output wire plus the cover of their difference, when that leaves fewer gates.

*Ripple adder: $S_1 = C_0 \oplus (A_1 \oplus B_1)$, 126 → 64 gates.*

**4. $X$-cancellation (Lemma 2)**

Two $X$ gates on the same wire with no control use of that wire between them cancel ($XX = I$): delete both.

*Full adder: 23 → 19 gates (the boxed pairs in Fig. 2).*

**5. Term ordering (Lemma 3)**

Groups commute, so reorder them to make more $X$ pairs meet and cancel; keep the best of the candidate circuits.

*Full adder: 19 → 14 gates, quantum cost 74.*

## tweedledum: Reed–Muller (ESOP)

**1. Reed–Muller expansion of each output**

Write each output as an XOR of AND-terms (an ESOP). Terms may overlap: an XOR of two terms that both contain a cell flips it twice, as intended.

**Two variants, each run separately:**

**PPRM** (positive-polarity Reed–Muller): every term uses only uncomplemented inputs; the form is unique.

*$Sum = A \oplus B \oplus C_{in}$, $Carry = AB \oplus AC_{in} \oplus BC_{in}$.*

**PKRM** (pseudo-Kronecker Reed–Muller): expands variable by variable, choosing the Shannon, positive-Davio or negative-Davio rule at each step, so terms can mix complemented and uncomplemented inputs.

*$Carry = B\bar{C}_{in} \oplus AC_{in} \oplus \bar{A}B$.*

**2. Each term → one multi-controlled $X$**

XOR-ing every term into the output wire computes it. For the comparison we rewrite negated controls as $X$-sandwiches and apply the same $X$-cancellation (step 4).

*Full adder: PPRM 6 gates (cost 18), PKRM 10 gates (cost 22).*

**Check: classical replay of every input row**

Each gate maps basis states to basis states, so running all $2^n$ input rows through the circuit proves it correct. Every circuit from both methods passes before it is counted.

**Multiple-control Toffoli (MCT) circuit on $n + m$ qubits**

Compared by gate count and quantum cost ($2^{k+1} - 3$ per $k$-control gate): Table I.

**Transpile to an IBM Heron device model (Qiskit, FakeTorino)**

Decompose into native CZ, $\sqrt{X}$, $R_z$ gates and route on the device's coupling map, with `qubits_initially_zero=False`; check again by simulating every input. Result: CZ count and depth (Table I).

**The key difference.** A Toffoli gate XORs into its output, so a K-map cell covered by two groups is flipped twice and cancels. *This work* forbids overlaps, which gives a short correctness proof (Theorems 1–2). *PPRM/PKRM* use overlapping XOR terms on purpose, which often needs far fewer gates (full adder: 6 or 10 vs. 14).